

BMS AUTOSAR ECU

System Requirements Specification

Model-Based Development with Simulink + EB Tresos

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Tools	Simulink R2024a + EB Tresos + STM32

1. Project Overview

This document defines the system requirements for a Battery Management System (BMS) ECU developed using AUTOSAR Classic architecture. The system uses Model-Based Development (MBD) in MATLAB/Simulink for algorithm design and EB Tresos for BSW configuration.

1.1 Project Goal

Design, simulate, and generate production-ready AUTOSAR C code for a BMS system that monitors, estimates, protects, and balances a lithium-ion battery pack in an automotive electric vehicle.

1.2 Tools & Technologies

Tool	Version	Purpose
MATLAB / Simulink	R2024a	BMS algorithm modeling and simulation
AUTOSAR Blockset	R2024a	AUTOSAR SWC design and ARXML generation
Embedded Coder	R2024a	MISRA-C compliant C code generation
Stateflow	R2024a	State machine logic (protection, balancing)
EB Tresos	Latest	BSW configuration (Com, Dem, NvM, Os)
STM32CubeIDE	Latest	Optional: integration on STM32 hardware
Cppcheck	Latest	MISRA-C static analysis
Git + GitHub	Latest	Version control and portfolio

2. System Architecture

2.1 AUTOSAR Layered Architecture

Layer	Components	Configured With
Application Layer	BMS_Monitor_SWC, BMS_Estimator_SWC, BMS_Protection_SWC, BMS_Balancing_SWC	Simulink + AUTOSAR Blockset
RTE	Inter-SWC communication, port mapping, runnable scheduling	Auto-generated from ARXML
Services Layer	Dem (faults), NvM (storage), Com (CAN), Os (scheduler)	EB Tresos
ECU Abstraction	CAN Interface, EEPROM abstraction, ADC abstraction	EB Tresos
MCAL	ADC driver, CAN driver, SPI driver, PWM driver	EB Tresos / STM32 HAL

2.2 Software Components (SWCs)

SWC Name	Responsibility	Runnable Period
BMS_Monitor_SWC	Read voltage, current, temperature from all cells	10 ms
BMS_Estimator_SWC	Calculate SOC (Coulomb Counting + Kalman), SOH, SOP	100 ms
BMS_Protection_SWC	Enforce voltage, current, temperature limits. Trigger DTCs	10 ms
BMS_Balancing_SWC	Passive cell balancing — bleed high cells via resistors	1000 ms
BMS_Communication_SWC	Send SOC, SOH, faults via CAN PDUs to vehicle network	20 ms

3. Functional Requirements

3.1 Monitoring Requirements

Req ID	Category	Requirement	Tool	Priority
FR-MON-01	Monitoring	System shall measure cell voltage for all 6 cells every 10 ms	Simulink ADC block	High
FR-MON-02	Monitoring	System shall measure pack current every 10 ms using Hall-effect sensor model	Simulink	High
FR-MON-03	Monitoring	System shall measure cell temperature for all 6 cells every 100 ms	Simulink	High
FR-MON-04	Monitoring	Measured values shall be validated for sensor plausibility before use	Simulink logic	Medium

3.2 SOC Estimation Requirements

Req ID	Category	Requirement	Tool	Priority
FR-SOC-01	Estimation	SOC shall be estimated using Coulomb Counting as primary method	Simulink Integrator	High
FR-SOC-02	Estimation	SOC shall be corrected using Extended Kalman Filter every 100 ms	Simulink EKF block	High
FR-SOC-03	Estimation	Initial SOC shall be determined from OCV-SOC lookup table at startup	Simulink Lookup Table	High
FR-SOC-04	Estimation	SOC estimation error shall not exceed 5% under normal conditions	Simulink Test	High
FR-SOC-05	Estimation	SOH shall be calculated based on capacity fade over charge cycles	Simulink	Medium
FR-SOC-06	Estimation	SOP (max available power) shall be estimated every 100 ms	Simulink	Medium

3.3 Protection Requirements

Req ID	Category	Requirement	Tool	Priority
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FR-PRO-01	Protection	System shall disconnect battery if any cell voltage exceeds 4.2V	Stateflow	Critical
FR-PRO-02	Protection	System shall disconnect battery if any cell voltage drops below 2.5V	Stateflow	Critical
FR-PRO-03	Protection	System shall limit charge current if any cell temperature exceeds 45 degrees C	Stateflow	Critical
FR-PRO-04	Protection	System shall disconnect battery if any cell temperature exceeds 60 degrees C	Stateflow	Critical
FR-PRO-05	Protection	System shall disconnect if discharge current exceeds 100A for more than 500 ms	Stateflow	Critical
FR-PRO-06	Protection	All protection events shall generate a DTC via AUTOSAR Dem module	EB Tresos Dem	High
FR-PRO-07	Protection	DTCs shall be stored in NvM and survive power cycle	EB Tresos NvM	High

3.4 Cell Balancing Requirements

Req ID	Category	Requirement	Tool	Priority
FR-BAL-01	Balancing	System shall implement passive balancing by bleeding energy from highest cell	Stateflow + PWM	High
FR-BAL-02	Balancing	Balancing shall activate when max-min cell voltage difference exceeds 50 mV	Simulink	High
FR-BAL-03	Balancing	Balancing shall deactivate when difference drops below 10 mV	Simulink	High
FR-BAL-04	Balancing	Balancing shall not activate during fast charging (current > 20A)	Stateflow	Medium

3.5 Communication Requirements

Req ID	Category	Requirement	Tool	Priority
FR-COM-01	Communication	System shall transmit SOC value on CAN ID 0x100 every 20 ms	EB Tresos Com	High
FR-COM-02	Communication	System shall transmit SOH value on CAN ID 0x101 every 1000 ms	EB Tresos Com	High

FR-COM-03	Communication	System shall transmit active DTC count on CAN ID 0x102 every 500 ms	EB Tresos Com + Dem	High
FR-COM-04	Communication	System shall transmit cell voltages (all 6) on CAN ID 0x110 every 100 ms	EB Tresos Com	Medium
FR-COM-05	Communication	System shall support UDS service 0x22 (Read Data by ID) for diagnostics	EB Tresos Dcm	Medium
FR-COM-06	Communication	System shall support UDS service 0x19 (Read DTC information)	EB Tresos Dcm	Medium

4. Non-Functional Requirements

Req ID	Category	Requirement	Tool	Priority
NFR-01	Safety	All generated C code shall comply with MISRA-C:2012 rules	Embedded Coder + Cppcheck	Critical
NFR-02	Safety	System shall implement ISO 26262 ASIL-B safety mechanisms for protection functions	Design + Review	Critical
NFR-03	Performance	BMS_Monitor_SWC runnable shall complete execution within 1 ms (10 ms period)	Simulink profiling	High
NFR-04	Performance	BMS_Estimator_SWC runnable shall complete within 5 ms (100 ms period)	Simulink profiling	High
NFR-05	Reliability	System shall handle sensor failure by switching to last known good value	Stateflow	High
NFR-06	Portability	Generated code shall be hardware-independent (no HAL calls in SWC)	AUTOSAR RTE	High
NFR-07	Traceability	All requirements shall be traceable to Simulink model blocks	Simulink Requirements Toolbox	Medium
NFR-08	Documentation	All functions shall be documented using Doxygen-compatible comments	Code generation config	Medium

5. BSW Configuration — EB Tresos

5.1 OS Configuration

Task Name	Period	Priority	SWC Runnable Called
Task_10ms	10 ms	High (8)	BMS_Monitor_Run, BMS_Protection_Run
Task_20ms	20 ms	Medium (6)	BMS_Communication_Run
Task_100ms	100 ms	Medium (5)	BMS_Estimator_Run
Task_1000ms	1000 ms	Low (3)	BMS_Balancing_Run
Task_Init	Startup only	Highest (10)	BMS_Init, NvM_ReadAll

5.2 Dem — DTC Definition

DTC ID	Name	Trigger	Severity
0xA10001	DTC_CELL_OVERVOLTAGE	Any cell > 4.2V	Critical
0xA10002	DTC_CELL_UNDERVOLTAGE	Any cell < 2.5V	Critical
0xA10003	DTC_OVERCURRENT_DISCHARGE	Current > 100A for 500ms	Critical
0xA10004	DTC_OVERTEMPERATURE	Any cell temp > 60 degrees C	Critical
0xA10005	DTC_SOC_ESTIMATION_ERROR	SOC error > 10%	Warning
0xA10006	DTC_CELL_IMBALANCE	Cell diff > 200 mV	Warning
0xA10007	DTC_SENSOR_PLAUSIBILITY	Sensor out of physical range	Warning

5.3 NvM — Non-Volatile Storage

Block Name	Size	Data Stored	Trigger
NvM_SOH_Block	4 bytes	Last known SOH value	On shutdown
NvM_DTC_Block	32 bytes	Active DTC list	On DTC change
NvM_CycleCount_Block	4 bytes	Charge cycle counter	On full charge

NvM_SOC_Block	4 bytes	Last SOC for next startup OCV init	On shutdown
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6. Simulink Model Requirements

6.1 Model Structure

Model File	Contains	Output
BMS_Monitor.slx	ADC input blocks, sensor validation, signal conditioning	Cell V, I, T signals
BMS_Estimator.slx	Coulomb counter, Kalman filter, OCV lookup table	SOC, SOH, SOP values
BMS_Protection.slx	Stateflow chart with all protection states	Relay control signals + DTC triggers
BMS_Balancing.slx	Balancing logic Stateflow, PWM output signals	Per-cell balancing enable flags
BMS_Communication.slx	CAN PDU packing, signal scaling, DTC aggregation	CAN signal values for Com layer
BMS_TestHarness.slx	Full system simulation with sensor signal generators	Scope outputs + test results

6.2 Stateflow Protection States

State	Entry Condition	Exit Condition	Action
NORMAL	All parameters in range	Any limit exceeded	Allow charge and discharge
WARNING	Temp > 45 degrees C OR Cell diff > 50mV	Parameters return to normal	Limit charge current to 50%
FAULT_VOLTAGE	Cell V > 4.2V or < 2.5V	Manual reset via UDS	Open main contactor, log DTC
FAULT_CURRENT	I > 100A for 500ms	Manual reset via UDS	Open main contactor, log DTC
FAULT_TEMP	Temp > 60 degrees C	Temp < 50 degrees C	Open main contactor, log DTC
SHUTDOWN	IgnOff or critical fault	IgnOn with no faults	NvM write, safe shutdown

7. Test & Verification Requirements

Test ID	Test Description	Tool	Pass Criteria
TEST-01	SOC accuracy test: ramp discharge 0-100% then recharge	Simulink Test	SOC error < 5%
TEST-02	Overvoltage protection: inject cell voltage > 4.2V	Simulink Test	DTC 0xA10001 active within 10ms
TEST-03	Undervoltage protection: inject cell voltage < 2.5V	Simulink Test	DTC 0xA10002 active within 10ms
TEST-04	Overtemperature: inject temperature > 60 degrees C	Simulink Test	Contactor open within 10ms
TEST-05	Cell balancing: inject cell voltage imbalance > 50mV	Simulink Test	Balancing activates within 1000ms
TEST-06	CAN PDU: verify SOC transmitted on 0x100 every 20ms	Simulink + Com	PDU rate matches exactly
TEST-07	NvM persistence: verify SOH survives simulated power cycle	EB Tresos NvM sim	SOH value preserved
TEST-08	MISRA compliance: run Cppcheck on all generated code	Cppcheck	Zero MISRA violations
TEST-09	Kalman filter convergence: verify SOC correction within 30s	Simulink Scope	Error < 2% after convergence
TEST-10	Model coverage: 100% decision coverage on protection logic	Simulink Coverage	Coverage = 100%

8. Project Deliverables

Deliverable	Description	Format
Simulink Models	5 SWC models + 1 test harness, fully parameterized	.slx files
Generated C Code	MISRA-compliant AUTOSAR C code for all 5 SWCs	.c / .h files
ARXML Files	Complete AUTOSAR software description for all SWCs	.arxml files
EB Tresos Project	Full BSW config: Os, Com, Dem, NvM, Dcm modules	Tresos workspace
Test Report	Simulink Test results with pass/fail for all 10 tests	.pdf report
GitHub Repository	Full project with README, architecture diagrams, docs	GitHub repo
This SRS Document	Complete system requirements specification	.pdf file

9. Project Folder Structure

```
BMS_AUTOSAR_Project/  
  Simulink/  
    BMS_Monitor.slx  
    BMS_Estimator.slx  
    BMS_Protection.slx  
    BMS_Balancing.slx  
    BMS_Communication.slx  
    BMS_TestHarness.slx  
  Generated_Code/  
    BMS_Monitor/   BMS_Estimator/   BMS_Protection/  
  ARXML/  
    BMS_Monitor.arxml ... SystemDescription.arxml  
  EB_Tresos/  
    Os_Config/  Com_Config/  Dem_Config/  NvM_Config/  
  Tests/  
    test_results.pdf  coverage_report.html  
  Docs/  
    BMS_SRS.pdf  Architecture.png  
  README.md
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